



Design, testing, and evaluation of protocol converter for DER networks.

Hounandan Ravichandran and Manimaran Govindarasu

Department of Electrical and Computer Engineering, Iowa State University, Ames, IA 50011

Email: nandan35@iastate.edu and gmani@iastate.edu



BACKGROUND

- ❖ DERs (rooftop solar, BESS, EVs) require utilities to monitor and control thousands of distributed devices.
- ❖ Four DER deployment scenarios[1]; Scenario 2 (gateway-based) is the most common pattern in North America.

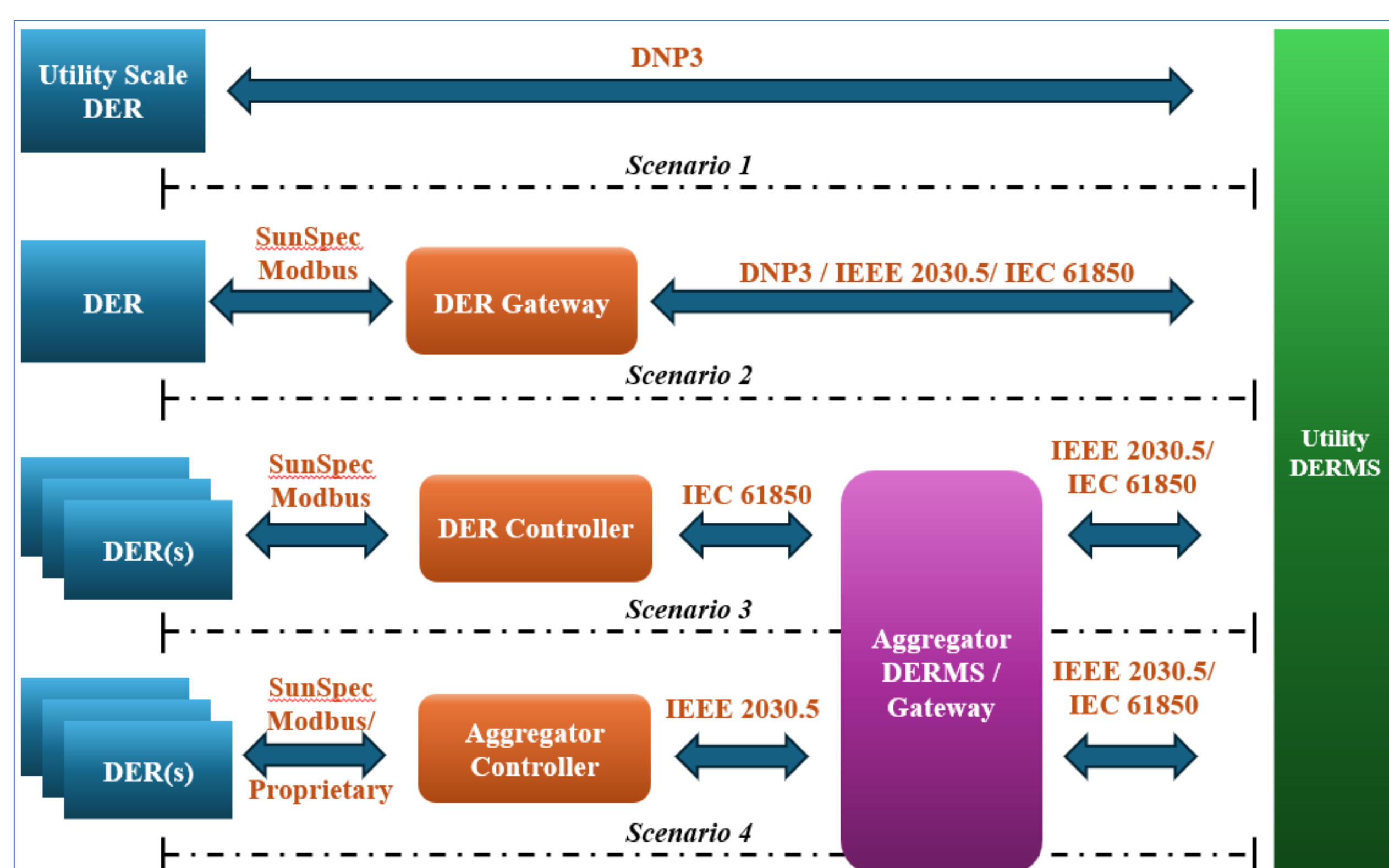


Figure 1: DER Deployment Scenarios

RESEARCH PROBLEM

- ❖ No openly available real-time converter bridges SunSpec Modbus and DNP3 at the Scenario 2 gateway.
- ❖ Both protocols run over plaintext TCP, exposing the gateway to reconnaissance and FDI attacks.

FUNCTIONAL VALIDATION: MONITORING AND CONTROL

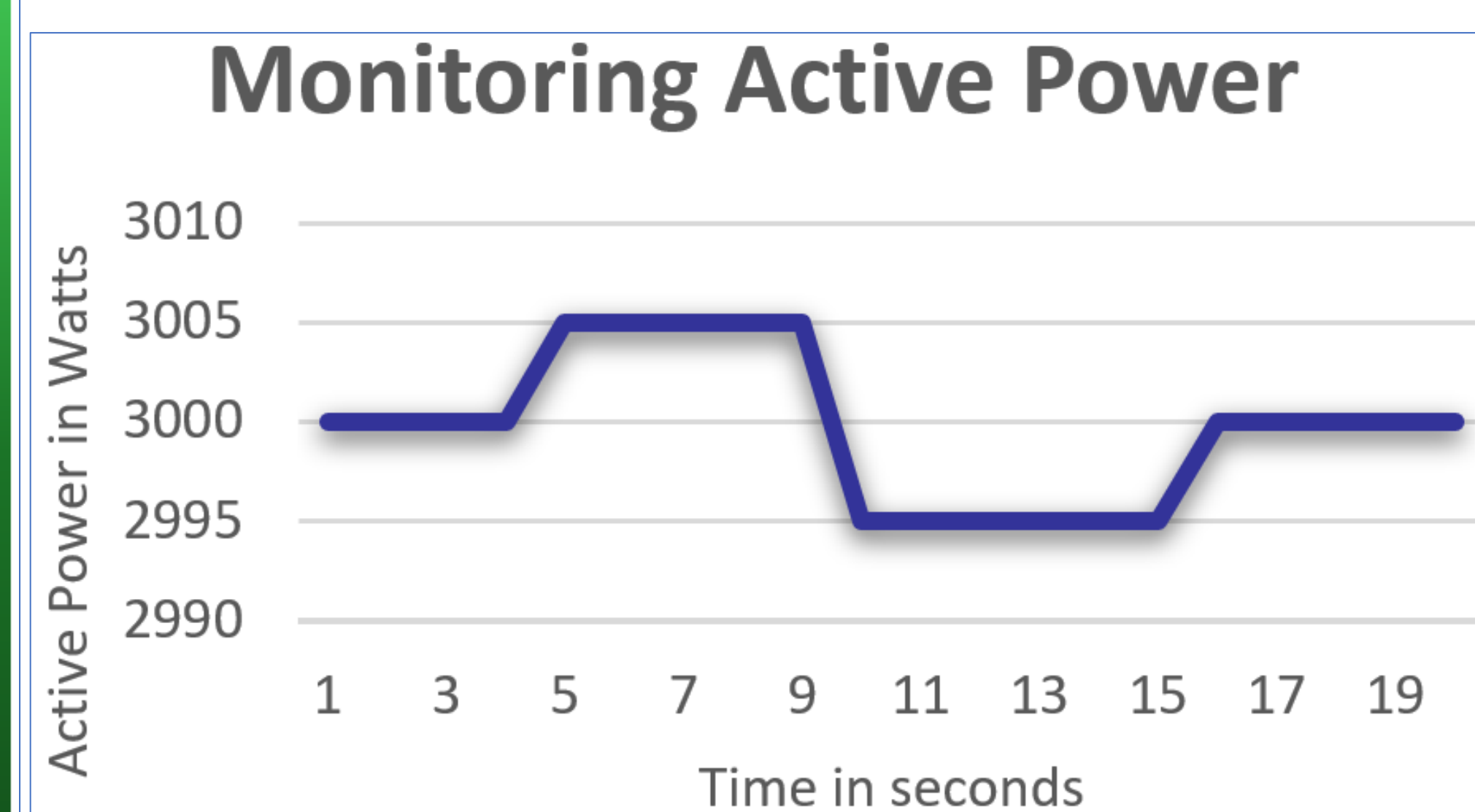


Figure 4: DER Data monitored by the DERMS

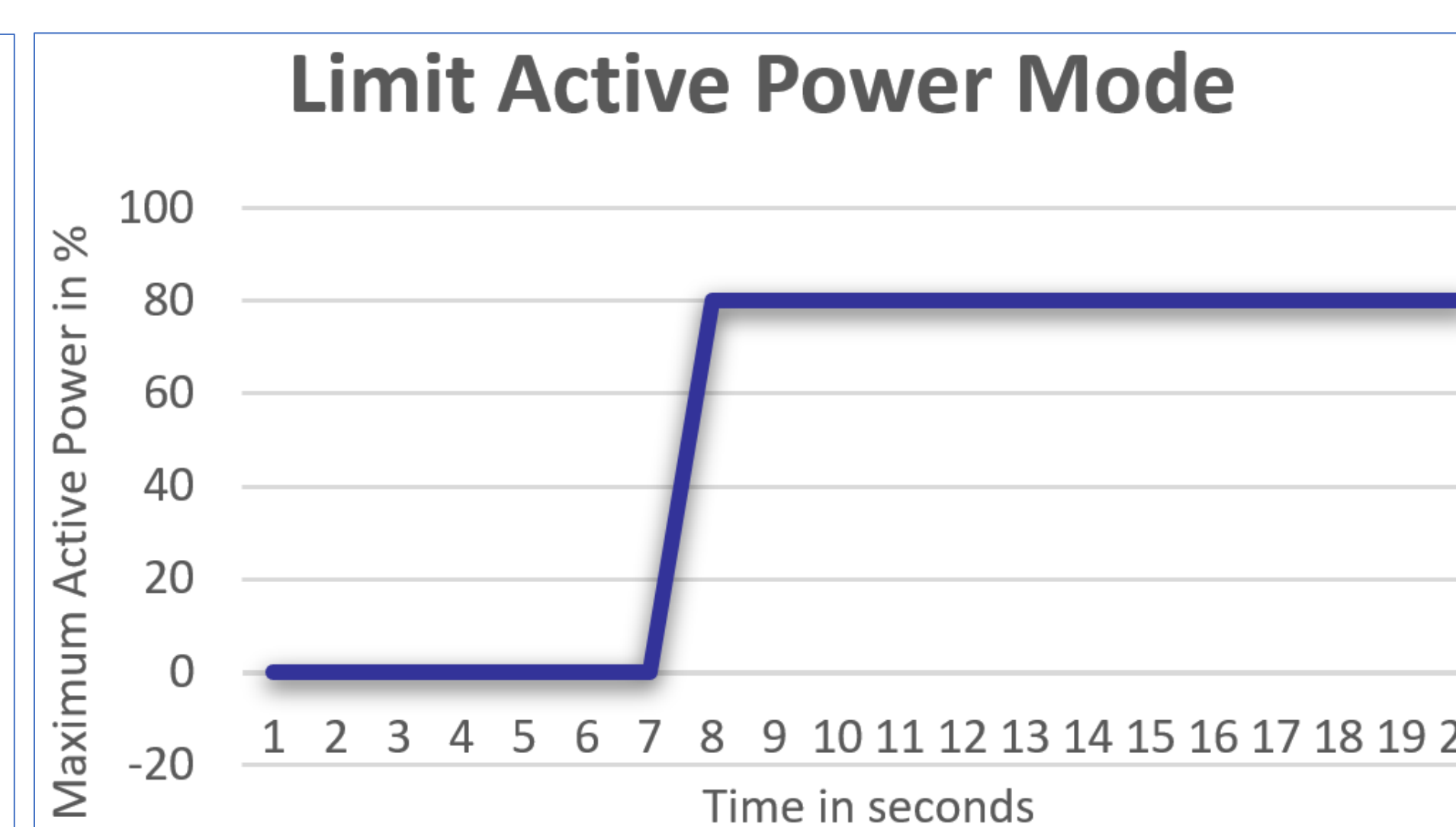


Figure 5: Received Control Setpoint in the DER

PROPOSED SOLUTION

- ❖ Two-endpoint architecture with three coordinated processes and a JSON-based internal data model.
- ❖ Standards-anchored mapping using IEEE 1547, SunSpec DER Information Model, and DNP Application Note 2018.

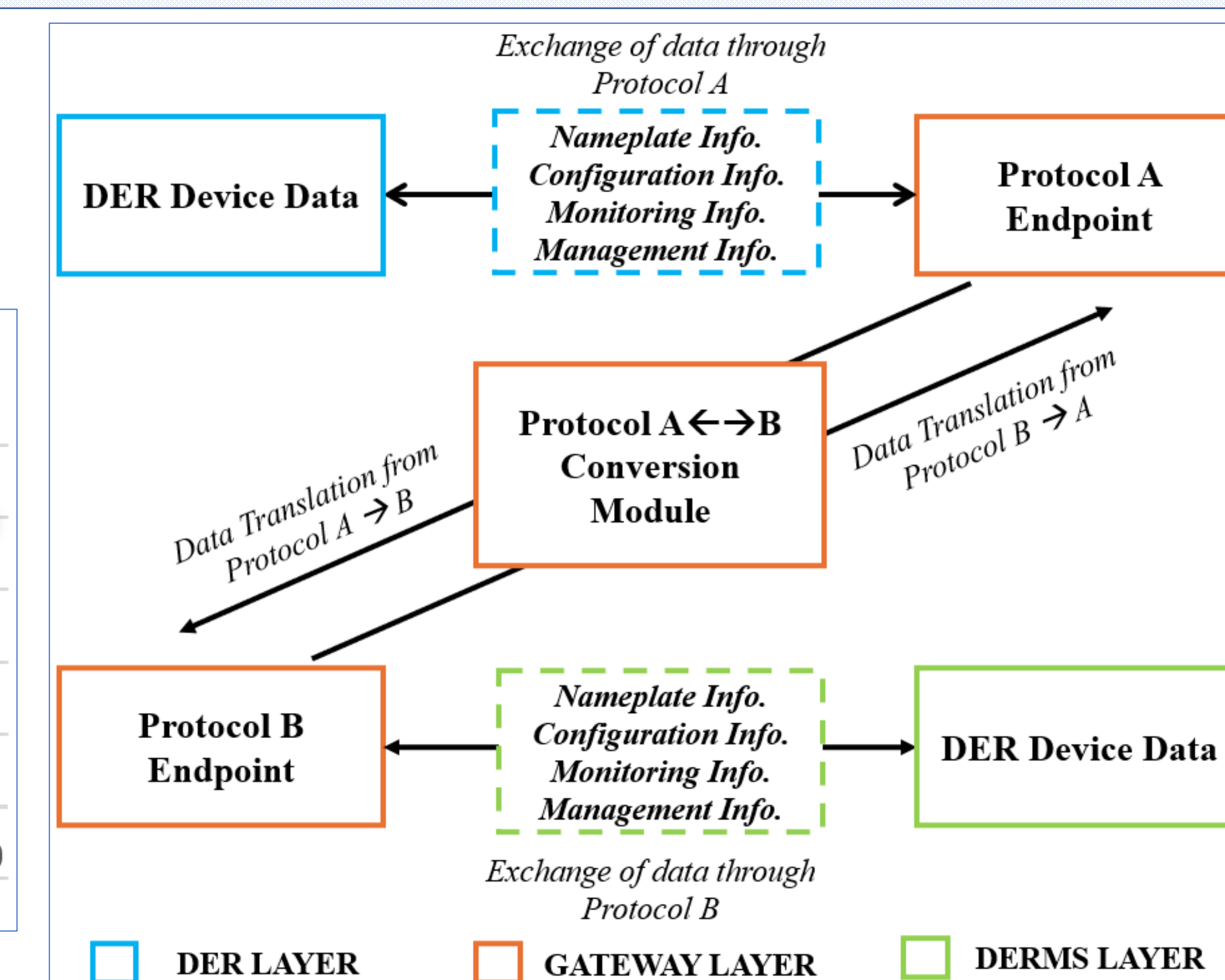


Figure 2: DER Data flow in Protocol Converter

EXPERIMENTAL ARCHITECTURE WITH TLS IN POWERCYBER TESTBED

- ❖ Deployed on PowerCyber testbed: SunSpec Modbus server (Linux VM), converter (NVIDIA Jetson Orin), DNP3master (DERMS).
- ❖ TLS 1.2 mTLS active on both interfaces, backed by a dedicated Root CA issuing per-role certificates.

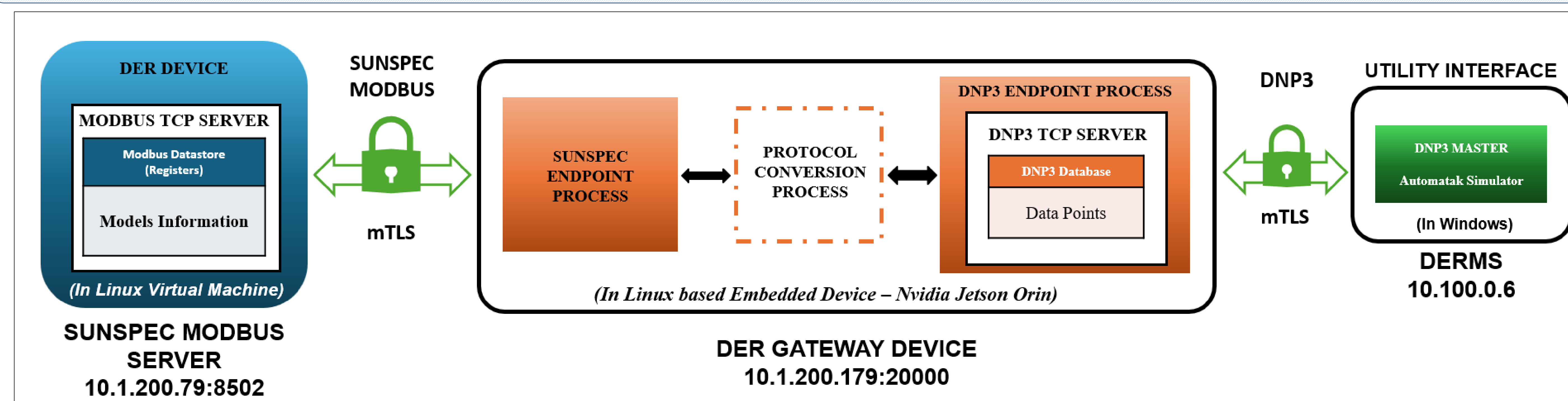


Figure 3: Secure, Real-Time, Bidirectional DER Network with Protocol Converter in PowerCyber Testbed

PERFORMANCE: BASELINE VS TLS LATENCY

Scenario	Baseline (ms)	TLS (ms)	Overhead (ms)
Single Read	11.22	34.17	+22.29
Single Write	11.19	51.51	+40.32
Multi Read	130.8	193.79	+63.71
Multi Write	160.53	407.06	+246.54

Table 1: End-to-End Latency for Baseline and TLS with overhead

CONCLUSION

- ❖ Interoperable[2] and secure SunSpec Modbus ↔ DNP3 converter for Scenario 2.
- ❖ TLS 1.2 mTLS provides CIA properties while meeting IEEE 1547 latency requirements.

[1] J. Ogle, M. Touhiduzzaman, Q. Nguyen, and N. Samaan, "CRest-VCT system integration framework," Pacific Northwest National Laboratory (PNNL), Tech. Rep. PNNL-29649, Jan 2020.

[2] H. Ravichandran, S. Bhattacharya, N. Saqib, and M. Govindarasu, "Real-time edge-based protocol converter and testbed-based evaluation for DER networks," in IEEE Power & Energy Society General Meeting (PESGM), Canada, 2026. (Accepted).